Clase21. Respuesta Multinomial

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1.- Data

La base de datos `nes96` , proveniente del National Election Study de 1996 (EE.UU.), contiene información de encuestas a individuos sobre comportamiento político y características socioeconómicas. Se trata de un estudio transversal con aproximadamente 900 observaciones, donde la unidad de análisis es el individuo.

Una de las variables de interés es `PID` , que representa la identificación política y se modela como una variable categórica nominal, ya que sus categorías son por ejemplo: Democrat, Republican, Independent y no poseen un orden natural. Como covariables, se incluyen variables como edad (age), nivel educacional (educ) e ingreso (income), entre otras.

Este conjunto de datos es particularmente adecuado para ilustrar modelos de regresión logística multinomial, permitiendo analizar cómo las características individuales influyen en la probabilidad de pertenecer a distintas categorías políticas, así como interpretar efectos mediante odds ratios y evaluar el ajuste del modelo mediante criterios como la deviance, test de razón de verosimilitud y pseudo R^2 .

```
# Paquetes necesarios
library(faraway)
library(nnet)

# Cargar datos
data(nes96)

# Revisar estructura
str(nes96)
```

```
## 'data.frame':    944 obs. of  10 variables:
##  $ popul : int  0 190 31 83 640 110 100 31 180 2800 ...
##  $ TVnews: int  7 1 7 4 7 3 7 1 7 0 ...
##  $ selfLR: Ord.factor w/ 7 levels "extLib"<"Lib"<..: 7 3 2 3 5 3 5 5 4 3 ...
##  $ ClinLR: Ord.factor w/ 7 levels "extLib"<"Lib"<..: 1 3 2 4 6 4 6 4 6 3 ...
##  $ DoleLR: Ord.factor w/ 7 levels "extLib"<"Lib"<..: 6 5 6 5 4 6 4 5 3 7 ...
##  $ PID   : Ord.factor w/ 7 levels "strDem"<"weakDem"<..: 7 2 2 2 1 2 2 5 4 1 ...
##  $ age   : int  36 20 24 28 68 21 77 21 31 39 ...
##  $ educ  : Ord.factor w/ 7 levels "MS"<"HSdrop"<..: 3 4 6 6 6 4 4 4 4 3 ...
##  $ income: Ord.factor w/ 24 levels "$3Kminus"<"$3K-$5K"<..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ vote  : Factor w/ 2 levels "Clinton","Dole": 2 1 1 1 1 1 1 1 1 1 ...
```

```
head(nes96)
```

```
##   popul TVnews selfLR ClinLR DoleLR     PID age  educ   income    vote
## 1     0      7 extCon extLib    Con  strRep  36    HS $3Kminus    Dole
## 2   190      1 sliLib sliLib sliCon weakDem  20  Coll $3Kminus Clinton
## 3    31      7    Lib    Lib    Con weakDem  24 BAdeg $3Kminus Clinton
## 4    83      4 sliLib    Mod sliCon weakDem  28 BAdeg $3Kminus Clinton
## 5   640      7 sliCon    Con    Mod  strDem  68 BAdeg $3Kminus Clinton
## 6   110      3 sliLib    Mod    Con weakDem  21  Coll $3Kminus Clinton
```

Variable respuesta nominal

En estos datos, podemos usar como respuesta la variable `PID` , que representa identificación política.

```
table(nes96$PID)
```

```
##
##  strDem weakDem  indDem  indind  indRep weakRep  strRep
##     200     180     108      37      94     150     175
```

La variable `PID` tiene categorías sin un orden natural claro para este ejemplo. Por eso se puede tratar como una respuesta nominal.

```
#nes96$PID <- factor(nes96$PID)

# Ver categoría de referencia
levels(nes96$PID)
```

```
## [1] "strDem"  "weakDem" "indDem"  "indind"  "indRep"  "weakRep" "strRep"
```

```
nes96$PID3 <- with(nes96,
  ifelse(PID %in% c("strDem","weakDem","indDem"), "Democrat",
  ifelse(PID %in% c("strRep","weakRep","indRep"), "Republican",
         "Independent"))
)

nes96$PID3 <- factor(nes96$PID3,
                     levels = c("Democrat","Republican","Independent"))

table(nes96$PID3)
```

```
##
##    Democrat  Republican Independent
##         488         419          37
```

```
levels(nes96$PID3)
```

```
## [1] "Democrat"    "Republican"  "Independent"
```

```
head(nes96)
```

```
##   popul TVnews selfLR ClinLR DoleLR     PID age  educ   income    vote
## 1     0      7 extCon extLib    Con  strRep  36    HS $3Kminus    Dole
## 2   190      1 sliLib sliLib sliCon weakDem  20  Coll $3Kminus Clinton
## 3    31      7    Lib    Lib    Con weakDem  24 BAdeg $3Kminus Clinton
## 4    83      4 sliLib    Mod sliCon weakDem  28 BAdeg $3Kminus Clinton
## 5   640      7 sliCon    Con    Mod  strDem  68 BAdeg $3Kminus Clinton
## 6   110      3 sliLib    Mod    Con weakDem  21  Coll $3Kminus Clinton
##         PID3
## 1 Republican
## 2   Democrat
## 3   Democrat
## 4   Democrat
## 5   Democrat
## 6   Democrat
```

La primera categoría (Democrat) será tomada por `multinom()` como categoría de referencia. Si queremos fijar una referencia específica:

```
#Fijamos como categoría de referencia "Democrat"
library(forcats)
nes96$PID3 <- fct_relevel(nes96$PID3, "Democrat")

#Otra forma de hacerlo
#levels(nes96$PID3)
#nes96$PID3 <- factor(nes96$PID3,
#                     levels = c("Democrat", "Republican", "Independent")) # Al dejar pri
mero "Democrat" lo estamos fijando
```

En este caso, todas las categorías se compararán contra “Democrat”.

Para ajustar el modelo multinomial usaremos algunas covariables de interés, entre ellas: edad, nivel de educación e ingresos.

```
mod_mult <- multinom(PID3 ~ age + educ + income, data = nes96)
```

```
## # weights:  96 (62 variable)
## initial  value 1037.090001
## iter  10 value 740.365331
## iter  20 value 731.176905
## iter  30 value 729.486079
## iter  40 value 729.263042
## iter  50 value 729.251742
## final  value 729.249881
## converged
```

```
summary(mod_mult)
```

```
## Call:
## multinom(formula = PID3 ~ age + educ + income, data = nes96)
##
## Coefficients:
##             (Intercept)         age   educ.L     educ.Q    educ.C     educ^4
## Republican    -1.087515 0.010134262 0.665725 -0.8301778 0.1326374 -0.0684107
## Independent   -9.078982 0.004992766 8.303808 -9.2035646 6.9116378 -3.8917077
##                 educ^5     educ^6  income.L  income.Q income.C   income^4
## Republican  -0.0761951 -0.1297865  2.156363 0.2550579 0.831164 -0.6887334
## Independent  1.9098777 -0.6021337 11.336121 3.5819946 3.259088  0.3553418
##               income^5 income^6   income^7  income^8    income^9 income^10
## Republican  -0.4721421 0.753171 -0.8124920 0.1469042  -0.2443035  0.389876
## Independent -9.7025063 4.301739  0.1224398 8.8999961 -10.5975349  1.107619
##              income^11 income^12  income^13    income^14   income^15 income^16
## Republican  -0.2905394  0.474651 -0.8640103   0.03044968  -0.2366145 0.6837177
## Independent -1.8704552 -1.639946 14.3976720 -10.74257256 -10.0599094 7.1245281
##              income^17 income^18  income^19  income^20  income^21  income^22
## Republican  -0.2626478 0.0101567  0.4233105  0.2626417 -0.3392655  0.2041223
## Independent  3.4561318 1.3316645 -3.6221436 -2.1093060 -1.8827473 -4.3309594
##              income^23
## Republican   0.3718466
## Independent -7.7627086
##
## Std. Errors:
##             (Intercept)        age    educ.L    educ.Q    educ.C    educ^4
## Republican    0.2660477 0.00444997 0.4473786 0.4081483 0.3404244 0.2722870
## Independent   0.5665001 0.01116194 0.4741304 0.4128782 0.4866360 0.4856151
##                educ^5    educ^6  income.L  income.Q  income.C  income^4
## Republican  0.222662 0.1803029 0.5334778 0.5153263 0.4848746 0.4263770
## Independent 0.505424 0.4442280 0.7539633 0.7859199 0.7337632 0.7801692
##              income^5  income^6  income^7  income^8  income^9 income^10
## Republican  0.4538262 0.4897840 0.5424622 0.5572881 0.5588640 0.5356943
## Independent 0.7376649 0.7469699 0.6435609 0.6874961 0.6265387 0.6860556
##             income^11 income^12 income^13 income^14 income^15 income^16
## Republican  0.4888327 0.4784185 0.4551349 0.4430193 0.4481017 0.4433534
## Independent 0.6804297 0.7321812 0.6358503 0.7134904 0.7567449 0.7388292
##             income^17 income^18 income^19 income^20 income^21 income^22
## Republican  0.4359244 0.4511064 0.4604477 0.4464682 0.4663985 0.4580317
## Independent 0.6970348 0.7147581 0.7094577 0.7273854 0.7902653 0.7112812
##             income^23
## Republican  0.4435862
## Independent 0.8488416
##
## Residual Deviance: 1458.5
## AIC: 1582.5
```

- `weights` : número de parámetros estimados (modelo grande → muchas dummies).
- `initial value` : valor inicial de la log-verosimilitud (o deviance).
- `iter` : evolución del algoritmo iterativo (tipo Newton-Raphson).
- `final value` : valor final optimizado.

- `converged` : Indica que el algoritmo sí encontró solución estable.

Comentario: El modelo inicio mal pero fue mejorando hasta que finalmente se estabilizó.

Respecto a los coeficientes, cada fila es una comparación contra la categoría de referencia.

Dado que nuestra referencia es `Democrat` , entonces:

$$\ln\left(\frac{P(\text{Republican})}{P(\text{Democrat})}\right)$$

$$\ln\left(\frac{P(\text{Independent})}{P(\text{Democrat})}\right)$$

Por ejemplo: `Republican  age = 0.0101`

Si la edad aumenta en 1 unidad, enonces el $\ln\left(\frac{P(\text{Republican})}{P(\text{Democrat})}\right)$ aumenta en 0.01. En términos de interpretación: A mayor edad es ligeramente más probable que un individuo se incline más por ser Republicano que Democrata.

Para las variables del tipo: educ.L, educ.Q, educ.C, `R` está usando contrastes polinomiales porque `educ` es factor ordenado (tiene niveles).

- `educ.L` : tendencia lineal.
- `educ.Q` : curvatura.
- `educ.C` : tercer orden.

Una forma de resolver la salida de variables que posiblemente no me generan un valor al modelo es escribiendo la linea a continuación:

```
options(contrasts = c("contr.treatment", "contr.treatment")) # elimina los contrastes po
linomiales
mod_mult <- multinom(PID3 ~ age + educ + income, data = nes96)
```

```
## # weights:  96 (62 variable)
## initial  value 1037.090001
## iter  10 value 742.881119
## iter  20 value 732.255441
## iter  30 value 729.993746
## iter  40 value 729.463662
## iter  50 value 729.257555
## iter  60 value 729.250200
## final  value 729.249879
## converged
```

```
summary(mod_mult)
```

```
## Call:
## multinom(formula = PID3 ~ age + educ + income, data = nes96)
##
## Coefficients:
##             (Intercept)         age educHSdrop    educHS  educColl educCCdeg
## Republican    -2.499502 0.010133009  0.7293974  1.068064  1.310622  1.128252
## Independent  -17.195741 0.004996971 14.9094824 14.691340 14.610315 14.270396
##             educBAdeg  educMAdeg income$3K-$5K income$5K-$7K income$7K-$9K
## Republican   1.190927  0.8455374     -1.394611     0.3588026    -0.3781631
## Independent 13.876481 13.8352763    -14.440778   -14.2200352    -0.3238413
##             income$9K-$10K income$10K-$11K income$11K-$12K income$12K-$13K
## Republican       0.6410272       0.7456588       0.7726017       0.5449322
## Independent      1.5480950     -13.2662179     -12.4274337       1.0673346
##             income$13K-$14K income$14K-$15K income$15K-$17K income$17K-$20K
## Republican        0.7638633      -0.3437016       0.3227693       0.4443445
## Independent       1.1158354       0.7413504     -14.0624515      -0.6204797
##             income$20K-$22K income$22K-$25K income$25K-$30K income$30K-$35K
## Republican        0.1898005      -0.1720251       0.5101885       0.6986622
## Independent      -0.5315948      -0.8400122     -17.4011059     -17.4944431
##             income$35K-$40K income$40K-$45K income$45K-$50K income$50K-$60K
## Republican         0.872929       0.8177070       0.4560368       1.1620101
## Independent        0.225551      -0.6240323      -0.1430574      -0.4526262
##             income$60K-$75K income$75K-$90K income$90K-$105K income$105Kplus
## Republican        1.2710690       1.2706677         1.533146       1.2972128
## Independent       0.9170345      -0.2568064         1.478595       0.9923377
##
## Std. Errors:
##             (Intercept)         age educHSdrop    educHS  educColl educCCdeg
## Republican    0.8972380 0.004449968  0.7567165 0.7095183 0.7137590 0.7341698
## Independent   0.9808313 0.011161808  0.6298970 0.3582310 0.3739295 0.5690456
##             educBAdeg educMAdeg income$3K-$5K income$5K-$7K income$7K-$9K
## Republican  0.7162574 0.7278052  1.173705e+00  7.427073e-01     0.7828994
## Independent 0.4385088 0.5407483  1.115914e-05  1.472628e-05     1.4901824
##             income$9K-$10K income$10K-$11K income$11K-$12K income$12K-$13K
## Republican       0.7620441    0.7738128867    8.176207e-01       0.7518183
## Independent      1.2577985    0.0000228256    3.953243e-05       1.3117760
##             income$13K-$14K income$14K-$15K income$15K-$17K income$17K-$20K
## Republican        0.8762948       0.8504544    6.964619e-01       0.6425688
## Independent       1.5505318       1.3194845    2.056943e-05       1.4791397
##             income$20K-$22K income$22K-$25K income$25K-$30K income$30K-$35K
## Republican        0.6881435       0.6491916    5.897337e-01    5.860832e-01
## Independent       1.4823587       1.4671269    1.832327e-06    1.511633e-06
##             income$35K-$40K income$40K-$45K income$45K-$50K income$50K-$60K
## Republican        0.5934037       0.6095632        0.608720       0.5708263
## Independent       1.2134821       1.4671132        1.279401       1.2758247
##             income$60K-$75K income$75K-$90K income$90K-$105K income$105Kplus
## Republican        0.5733107       0.6067882         0.625182       0.5978488
## Independent       1.1319956       1.4767153         1.204565       1.2059825
##
## Residual Deviance: 1458.5
## AIC: 1582.5
```

```
coeficientes <- summary(mod_mult)$coefficients
errores <- summary(mod_mult)$standard.errors

coeficientes
```

```
##             (Intercept)         age educHSdrop    educHS  educColl educCCdeg
## Republican    -2.499502 0.010133009  0.7293974  1.068064  1.310622  1.128252
## Independent  -17.195741 0.004996971 14.9094824 14.691340 14.610315 14.270396
##             educBAdeg  educMAdeg income$3K-$5K income$5K-$7K income$7K-$9K
## Republican   1.190927  0.8455374     -1.394611     0.3588026    -0.3781631
## Independent 13.876481 13.8352763    -14.440778   -14.2200352    -0.3238413
##             income$9K-$10K income$10K-$11K income$11K-$12K income$12K-$13K
## Republican       0.6410272       0.7456588       0.7726017       0.5449322
## Independent      1.5480950     -13.2662179     -12.4274337       1.0673346
##             income$13K-$14K income$14K-$15K income$15K-$17K income$17K-$20K
## Republican        0.7638633      -0.3437016       0.3227693       0.4443445
## Independent       1.1158354       0.7413504     -14.0624515      -0.6204797
##             income$20K-$22K income$22K-$25K income$25K-$30K income$30K-$35K
## Republican        0.1898005      -0.1720251       0.5101885       0.6986622
## Independent      -0.5315948      -0.8400122     -17.4011059     -17.4944431
##             income$35K-$40K income$40K-$45K income$45K-$50K income$50K-$60K
## Republican         0.872929       0.8177070       0.4560368       1.1620101
## Independent        0.225551      -0.6240323      -0.1430574      -0.4526262
##             income$60K-$75K income$75K-$90K income$90K-$105K income$105Kplus
## Republican        1.2710690       1.2706677         1.533146       1.2972128
## Independent       0.9170345      -0.2568064         1.478595       0.9923377
```

```
errores
```

```
##             (Intercept)         age educHSdrop    educHS  educColl educCCdeg
## Republican    0.8972380 0.004449968  0.7567165 0.7095183 0.7137590 0.7341698
## Independent   0.9808313 0.011161808  0.6298970 0.3582310 0.3739295 0.5690456
##             educBAdeg educMAdeg income$3K-$5K income$5K-$7K income$7K-$9K
## Republican  0.7162574 0.7278052  1.173705e+00  7.427073e-01     0.7828994
## Independent 0.4385088 0.5407483  1.115914e-05  1.472628e-05     1.4901824
##             income$9K-$10K income$10K-$11K income$11K-$12K income$12K-$13K
## Republican       0.7620441    0.7738128867    8.176207e-01       0.7518183
## Independent      1.2577985    0.0000228256    3.953243e-05       1.3117760
##             income$13K-$14K income$14K-$15K income$15K-$17K income$17K-$20K
## Republican        0.8762948       0.8504544    6.964619e-01       0.6425688
## Independent       1.5505318       1.3194845    2.056943e-05       1.4791397
##             income$20K-$22K income$22K-$25K income$25K-$30K income$30K-$35K
## Republican        0.6881435       0.6491916    5.897337e-01    5.860832e-01
## Independent       1.4823587       1.4671269    1.832327e-06    1.511633e-06
##             income$35K-$40K income$40K-$45K income$45K-$50K income$50K-$60K
## Republican        0.5934037       0.6095632        0.608720       0.5708263
## Independent       1.2134821       1.4671132        1.279401       1.2758247
##             income$60K-$75K income$75K-$90K income$90K-$105K income$105Kplus
## Republican        0.5733107       0.6067882         0.625182       0.5978488
## Independent       1.1319956       1.4767153         1.204565       1.2059825
```

Las probabilidades para nuestro modelos multinomial vienen dadas por:

$$\widehat{p}_j = \frac{\exp\left(x_j^\top \widehat{\beta}_j\right)}{1 + \sum_{j=2}^{J} \exp\left(x_j^\top \widehat{\beta}_j\right)} \quad j = 2, \dots, J:$$

Y la categoría de referncia:

$$\widehat{p}_1 = \frac{1}{1 + \sum_{j=2}^{J} \exp\left(x_j^\top \widehat{\beta}_j\right)}$$

```
prob_hat <- fitted(mod_mult)

head(prob_hat)
```

```
##    Democrat Republican Independent
## 1 0.6934856  0.2386698  0.06784461
## 2 0.6866863  0.2561230  0.05719066
## 3 0.7218207  0.2487366  0.02944263
## 4 0.7140495  0.2562368  0.02971366
## 5 0.6293197  0.3386983  0.03198199
## 6 0.6847041  0.2579846  0.05731124
```

Por lo tanto; para cada individuo i:

$$\widehat{p}_{i,1} = \text{prob_hat}[i, \text{Democrat}] = \widehat{p}_1$$

$$\widehat{p}_{i,2} = \text{prob_hat}[i, \text{Republican}] = \widehat{p}_2$$

$$\widehat{p}_{i,3} = \text{prob_hat}[i, \text{Independent}] = \widehat{p}_3$$

Y se cumple:

$$\widehat{p}_1 + \widehat{p}_2 + \widehat{p}_3 = 1$$

```
head(rowSums(prob_hat))
```

```
## 1 2 3 4 5 6
## 1 1 1 1 1 1
```

```
prob_hat[1, ]
```

```
##    Democrat  Republican Independent
##  0.69348559  0.23866979  0.06784461
```

Interpretación:

- 69% probabilidad de ser Democrat
- 24% Republican
- 7% Independent

Nuestro modelo está asignando probabilidades pero no clasifica.

```
pred_clase <- predict(mod_mult, type = "class")

table(Observado = nes96$PID3, Predicho = pred_clase)
```

```
##             Predicho
## Observado     Democrat Republican Independent
##   Democrat         322        166           0
##   Republican       197        222           0
##   Independent       17         20           0
```

Interpretación:

- `Democrat`
  - 322 bien clasificados.
  - 166 mal clasificados como Republican.
- `Republican`
  - 222 bien clasificados.
  - 197 mal clasificados como Democrat.
- `Independent`

- 0 correctamente clasificados. Todos se asignan a Democrat o Republican

```
summary(prob_hat)
```

```
##     Democrat        Republican       Independent
##  Min.   :0.2425   Min.   :0.05209   Min.   :0.00000
##  1st Qu.:0.4119   1st Qu.:0.36411   1st Qu.:0.01209
##  Median :0.5059   Median :0.46485   Median :0.02867
##  Mean   :0.5170   Mean   :0.44385   Mean   :0.03919
##  3rd Qu.:0.6133   3rd Qu.:0.54326   3rd Qu.:0.05287
##  Max.   :0.9479   Max.   :0.67901   Max.   :0.29059
```

Explicación de lo que observamos para la clase `Independent` . El modelo sí calcula probabilidades para Independent pero nunca son lo suficientemente grandes como para ganarle a las otras categorías por esa razón no clasifica ningún individuo en esa clase.

```
mean(pred_clase == nes96$PID3)
```

```
## [1] 0.5762712
```

Esta calculando la exactitud del modelo. Esto quiere decir que el modelo está claculando el 57% de los casos.

```
#install.packages("caret")
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
##
## Attaching package: 'lattice'
```

```
## The following object is masked from 'package:faraway':
##
##     melanoma
```

```
# Asegurarse que observado y predicho tengan los mismos niveles
pred_clase <- factor(pred_clase, levels = levels(nes96$PID3))

# Matriz de confusión y medidas
confusionMatrix(
  data = pred_clase,
  reference = nes96$PID3
)
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction    Democrat Republican Independent
##   Democrat         322        197          17
##   Republican       166        222          20
##   Independent        0          0           0
##
## Overall Statistics
##
##                Accuracy : 0.5763
##                  95% CI : (0.544, 0.608)
##     No Information Rate : 0.5169
##     P-Value [Acc > NIR] : 0.0001455
##
##                   Kappa : 0.1767
##
##  Mcnemar's Test P-Value : 1.266e-08
##
## Statistics by Class:
##
##                      Class: Democrat Class: Republican Class: Independent
## Sensitivity                   0.6598            0.5298            0.00000
## Specificity                   0.5307            0.6457            1.00000
## Pos Pred Value                0.6007            0.5441                NaN
## Neg Pred Value                0.5931            0.6325            0.96081
## Prevalence                    0.5169            0.4439            0.03919
## Detection Rate                0.3411            0.2352            0.00000
## Detection Prevalence          0.5678            0.4322            0.00000
## Balanced Accuracy             0.5953            0.5878            0.50000
```

Bondad de Ajuste del modelo

La deviance es

$$D = -2\left(\ell(\hat{\beta}) - \ell(\hat{\beta}_{max})\right) \sim \chi^2_{m-p,\nu}$$

La Deviance constituye la base de la mayoría de las pruebas de hipótesis de los modelos lineales generalizados.

```
mod_null <- multinom(PID3 ~ 1, data = nes96)
```

```
## # weights:  6 (2 variable)
## initial  value 1037.090001
## final  value 782.173305
## converged
```

```
logLik(mod_null)
```

```
## 'log Lik.' -782.1733 (df=2)
```

```
logLik(mod_mult)
```

```
## 'log Lik.' -729.2499 (df=62)
```

```
LR <- 2 * (as.numeric(logLik(mod_mult)) - as.numeric(logLik(mod_null)))
LR
```

```
## [1] 105.8469
```

```
gl <- attr(logLik(mod_mult), "df") - attr(logLik(mod_null), "df")
gl
```

```
## [1] 60
```

```
pvalor <- pchisq(LR, df = gl, lower.tail = FALSE)
pvalor
```

```
## [1] 0.0002405285
```

```
if(pvalor < 0.05){
  "El modelo con covariables mejora significativamente respecto al modelo nulo."
} else {
  "No hay evidencia suficiente de que las covariables mejoren el modelo nulo."
}
```

```
## [1] "El modelo con covariables mejora significativamente respecto al modelo nulo."
```

```
deviance(mod_mult)
```

```
## [1] 1458.5
```

```
deviance(mod_null)
```

```
## [1] 1564.347
```

```
deviance(mod_null) - deviance(mod_mult)
```

```
## [1] 105.8469
```